Battery State-of-Health

Degradation Models — Details & Accuracy

Mahindra · Euler · Bajaj | 22 Jun 2026

Scope

Per-OEM SoH forecasting models, their architecture and hyperparameters, and their out-of-sample accuracy measured by leave-one-vehicle-out (LOVO) backtest. Accuracy is reported separately for genuinely-degrading vs flat vehicles — the degrading cohort is the one that matters for warranty and RUL decisions.

Methodology

How the models are built and how accuracy is measured

SoH target (what we predict)

- Mahindra: coulomb-counting SoH from intellicar pack current (ΔSoC-weighted pooled).
- Euler: BMS remaining-capacity SoH at high SoC, isotonic-decreasing fit (validated method).
- Bajaj: BMS reported SoH (clean, monotone-enforced). All targets are monthly, per vehicle.

Model families

- Rate model: predict monthly ΔSoH (loss) from operating conditions + curvature, roll forward.
- Trajectory model (Euler primary): predict cumulative loss vs horizon, with P10/P50/P90 bands.
- A flat-vs-degrading gate prevents the model manufacturing loss on genuinely flat vehicles.

Validation — Leave-One-Vehicle-Out (LOVO)

- Hold out one whole vehicle at a time; train on all others (no leakage across vehicles).
- Within the held-out vehicle: give the model the first 60% of months, forecast the last 40%.
- Report RMSE/MAE split into degrading (tail loses ≥2 pp) vs flat vehicles.
- Baselines to beat: persistence (last SoH flat) and a √t-trend fit.
- Euler also reports P10–P90 band coverage (target ≈ 0.80).

How to read the accuracy

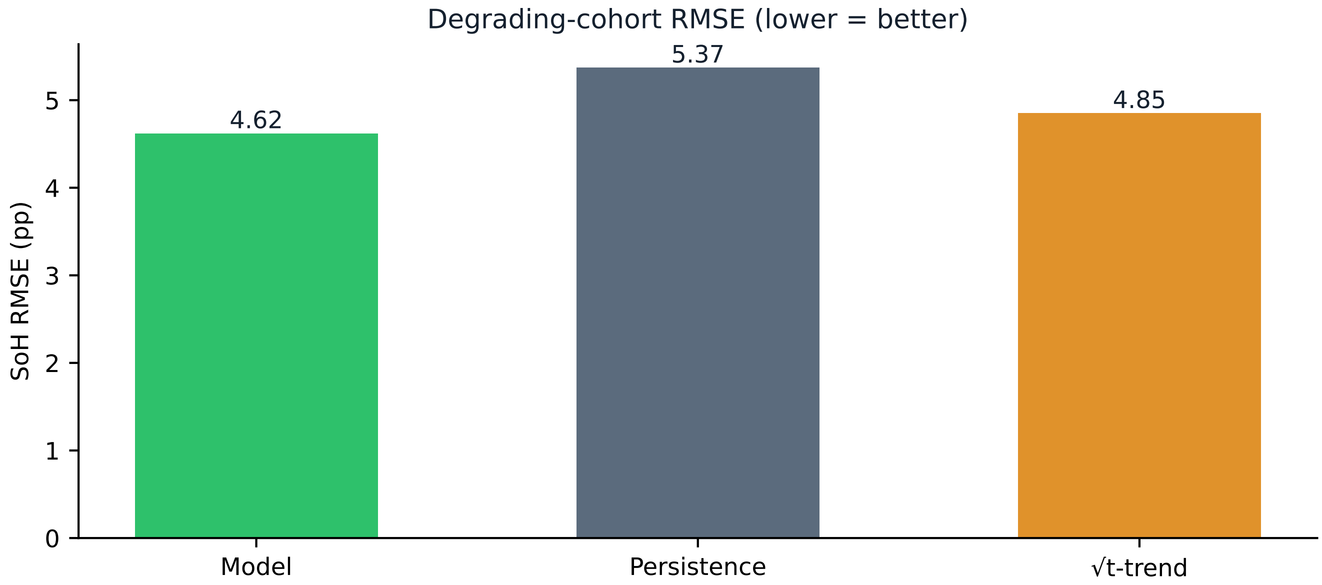
Lower RMSE/MAE (in SoH percentage-points) is better. On flat vehicles persistence is hard to beat (SoH barely moves); the test that matters is the degrading cohort, where the conditioned model must beat persistence and trend.

Mahindra — model & LOVO accuracy

Condition-aware rate model — LightGBM quantile (q10/q50/q90)

SoH target	Coulomb-counting SoH (intellicar current)
Hyperparameters	500 trees · lr 0.03 · num_leaves 15 · min_child 20 · flat-vs-degrading gate
Features	STATE(soh,age,cum_ah,cum_km,odo) + STRESS(current/SoC/volt/temp/km) + curvature(1/√age, deficit)
Cohort (LOVO)	84 vehicles · 28 degrading · 56 flat

Cohort	Model RMSE	Model MAE	Persist. RMSE	Trend RMSE
overall	3.15	1.77	3.22	3.73
degrading	**4.62**	2.89	5.37	4.85
flat	1.94	1.15	0.54	2.94



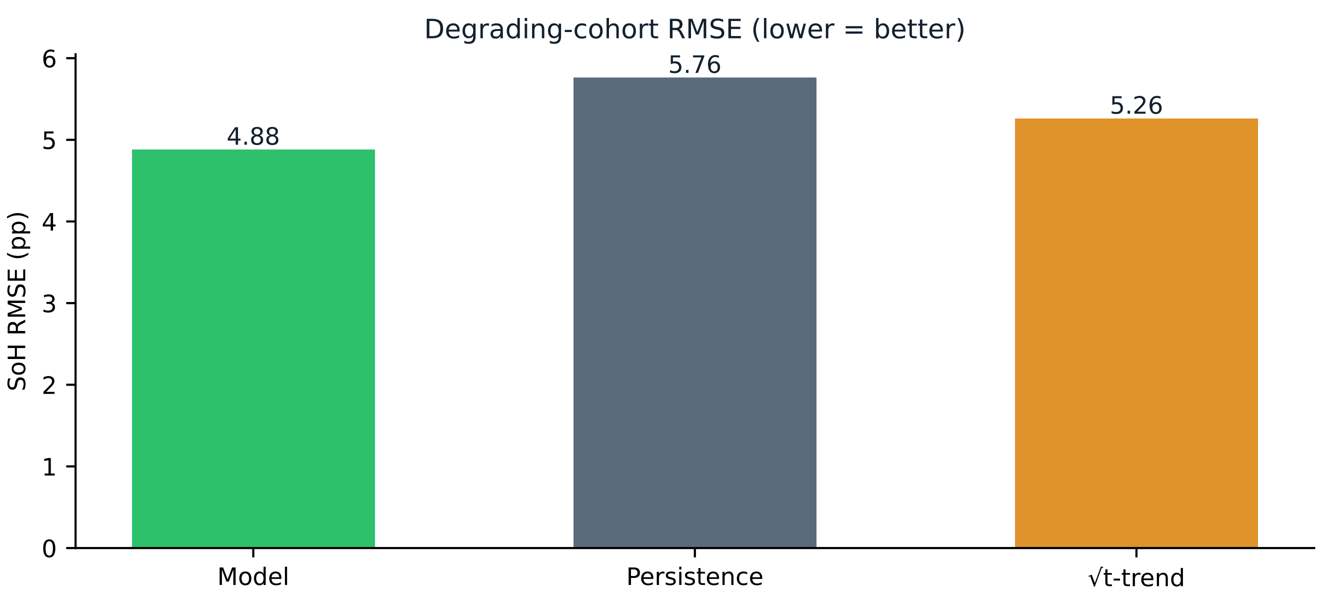
Conditioned model beats both baselines on the degrading cohort.

Euler — model & LOVO accuracy

Trajectory model — LightGBM quantile P50 + own-slope blend, √-horizon P10/P90 bands

SoH target	BMS remaining-capacity SoH (high-SoC, isotonic)
Hyperparameters	400 trees · lr 0.03 · num_leaves 15 · reg_lambda 2.0 · flat-pin
Features	anchor state + recent stress (TRAJ_STRESS: throughput/temp/DoD/SoC) + Δage, √Δage
Cohort (LOVO)	80 vehicles · 34 degrading · 46 flat

Cohort	Model RMSE	Model MAE	Persist. RMSE	Trend RMSE
overall	3.60	1.99	4.01	3.78
degrading	**4.88**	3.14	5.76	5.26
flat	1.67	0.92	0.44	1.37



Conditioned model beats both baselines on the degrading cohort.

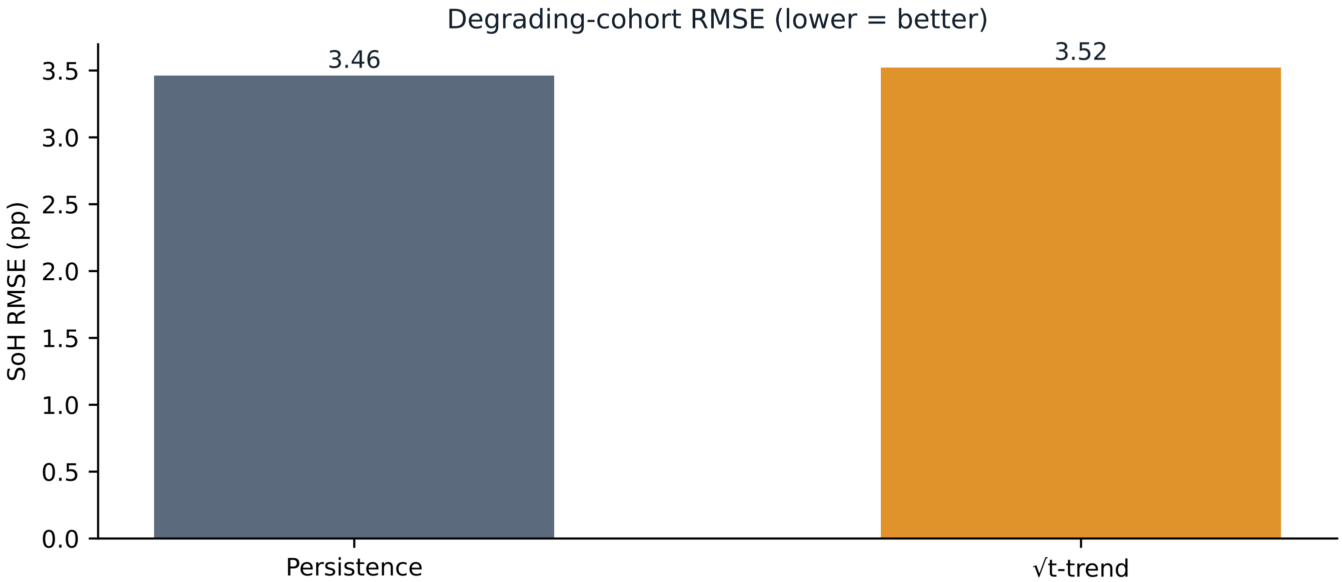
P10–P90 band coverage (target 0.80): overall 0.80, degrading 0.70, flat 0.89.

Bajaj — model & LOVO accuracy

No conditioned forecaster yet — √t-trend / persistence baseline (10-mo window)

SoH target	BMS reported SoH (essBmsSohcEstPercValue, cleaned)
Hyperparameters	n/a (feature table built; conditioned model is the next step)
Features	cycle count / SoC dwell / pack & ambient temp / drive-eff / odometer
Cohort (LOVO)	16 vehicles · 12 degrading · 4 flat

Cohort	Model RMSE	Model MAE	Persist. RMSE	Trend RMSE
overall	—	—	3.00	3.06
degrading	—	—	3.46	3.52
flat	—	—	0.76	1.00



Baselines only — conditioned Bajaj forecaster is the next step.

Summary

Headline accuracy and caveats

Headline (degrading-cohort RMSE, SoH pp)

- Mahindra: model 4.62 vs persistence 5.37 vs trend 4.85 — model wins (−14% vs persistence).
- Euler: model 4.88 vs persistence 5.76 vs trend 5.26 — model wins (−15% vs persistence).
- Bajaj: baselines ~3.0–3.5 (no conditioned model yet); short 10-mo window.

What the models are good / not good at

- Good: ordering and tracking genuine decliners — the warranty/RUL-relevant cohort.
- Flat vehicles are routed toward persistence by the gate (small added error vs persistence).
- Euler P10–P90 bands are calibrated to ~80% coverage overall (70% on steep degraders).

Caveats

- Small cohorts (tens of vehicles); LOVO RMSE is the production error bar — re-backtest as data grows.
- Bajaj needs a conditioned forecaster built (reuse src/model.py / euler_model.py).
- Accuracy is on the observed window; true multi-year extrapolation is unvalidated until vehicles age.